Prabhavit Gupta

LLM Engineer | Applied Generative AI, Python, TypeScript/Node.js, RAG & Vector Search

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Executive Summary

- **Delivery-Focused LLM Engineering for Production Applications:** Software engineer with 3 years commercial experience integrating pragmatic generative AI capabilities into existing production codebases (Python, Node.js/TypeScript). Experienced in building real-world AI features: document Q&A, semantic search, automated summarization, intent classification, and structured data extraction.
- **Production RAG Pipelines, Vector Search & Optimization:** Hands-on builder of robust Retrieval-Augmented Generation (RAG) architectures utilizing PostgreSQL pgvector, Qdrant, and Pinecone. Deep understanding of hybrid retrieval (BM25 + dense embeddings), reranking, and cost/latency constraints, optimizing token budgets and caching to maintain high throughput and reliability.
- **Autonomous Tool Integration & B2B Contracting:** Practical implementation of tool calling, structured outputs (Pydantic), and the Model Context Protocol (MCP) to safely connect LLMs (Anthropic Claude 3.5, OpenAI GPT-4o) with APIs and relational databases. Registered EU tax entity (Italian Partita IVA) fully prepared for immediate B2B freelance contracting.

Technical Skills

- **Languages & Core Runtimes:** Python (FastAPI, AsyncIO, PyDantic, PyTest), JavaScript & TypeScript (Node.js, Bun, React/Next.js), SQL (PostgreSQL), Bash.
- **LLM Engineering & Frameworks:** OpenAI API, Anthropic Claude API, LangChain, LlamaIndex, LangGraph, Model Context Protocol (MCP), Function/Tool Calling, Structured Outputs, Prompt Engineering & Iterative Refinement.
- **Vector Databases & Search:** PostgreSQL (pgvector extension, HNSW indexing), Qdrant, Pinecone, ChromaDB, Hybrid Search (Dense + BM25 Sparse), Semantic Reranking (Cohere), Semantic Caching (Redis).
- **Data Pipelines & Infrastructure:** Ingestion Pipelines for Unstructured Data (PDFs, Markdown, JSON), Docker, RESTful APIs, Git, GitHub Actions CI/CD, AWS (S3, Lambda).
- **Evals & Observability:** RAGAS Framework, Precision/Recall Benchmarking, Hallucination Prevention, Token Latency & Cost Optimization.

Professional Experience

LLM Solutions Engineer & Consultant • *Independent / B2B AI Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Partnered with European tech companies and startups to design, implement, and integrate custom LLM features into existing Python and Node.js codebases under B2B contracts.
- Built resilient multi-tenant RAG systems on PostgreSQL with pgvector, incorporating semantic chunking, Cohere reranking, and Redis semantic cache to achieve sub-second query latency and reduce inference costs by 35%.
- Integrated Anthropic Claude and OpenAI APIs into client workflows, enforcing strict JSON output schemas via Pydantic to ensure deterministic integration with downstream backend services.
- Engineered custom connectors using the Model Context Protocol (MCP), enabling LLM agents to perform auditable queries against client databases and internal SaaS tools.
- Established automated prompt regression suites and evaluation harnesses (measuring faithfulness and context relevance), ensuring model changes never degraded customer experience.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed and deployed scalable microservices in Python (FastAPI) and TypeScript (Node.js) for automated document parsing, classification, and text extraction.
- Integrated vector search capabilities into core products, designing indexing workflows and asynchronous background ingestion jobs using Redis queues.
- Optimized PostgreSQL database schemas and query plans for high-throughput relational and vector storage, supporting growing customer data volume.
- Collaborated directly with product teams to scope AI enhancements, translating ambiguous business requests into reliable, tested software deliverables.

Key Engineering Projects

- **Production Document Intelligence Engine (Python + pgvector):** Lightweight RAG engine combining dense vector embeddings and BM25 sparse retrieval over thousands of technical documentation files, complete with meta-

data filtering and hallucination guardrails.

- **Full-Stack LLM Agent Framework (TypeScript / Node.js):** Autonomous agent platform implementing tool execution against external APIs and databases, featuring state preservation, streaming output (SSE), and token consumption metrics.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous education in formal logic, decision theory, analytical reasoning, and complex systems modeling.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Contracting & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted right to work across the EU without visa sponsorship; Registered Italian Partita IVA ready for immediate B2B invoicing.